AN INTELLIGENT AI-BASED FRAMEWORK FOR REAL-TIME CIVIC ISSUE REPORTING AND RESOLUTION

A PROJECT REPORT

Submitted to

Amrita Vishwa Vidyapeetham

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

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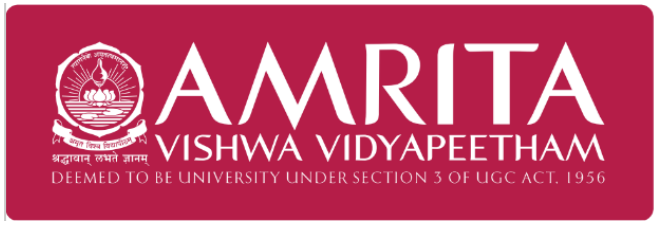
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ABSTRACT

Multilingual public issue classification still remains a tough task for smart city governance systems, especially in languages rich countries such as India, where citizens often raise issues in English, Hindi, and code mixed Hinglish. Current transformer-based multilingual approaches such as mBERT, MuRIL, IndicBERT and XLM-RoBERTa, primarily built for general purpose multilingual encoding, are neither trained for understanding domain-specific civic complaint text, nor for predicting hierarchical categories and handling significant class imbalances in actual civic grievance data. In this work, we propose HiENBERT (Hindi-English BERT), a hierarchical multi-task transformer architecture implemented in two versions – HiENBERT-Indic and HiENBERT-XLM, with IndicBERT and XLM-RoBERTa backbones. Our framework predicts primary category (8 classes), sub-category (24 classes) and severity (3 tiers) simultaneously using a shared representation layer and task-specific prediction heads, coupled with a sentiment augmented severity head, Focal Loss, class weighting, and weighted random sampling for handling class imbalances. We also introduce a new multilingual benchmarking dataset CivicComp (60,708 samples in English, Hindi and Hinglish) derived from the BBMP OpenCity platform, augmented with 15,000 synthetic complaints generated via five Large Language Models. Experimental results show that combining original and synthetic data yields up to 28.4% relative improvement in Macro-F1 over the imbalanced baseline, with HiENBERT-XLM achieving 93.28% accuracy and 93.42% Macro-F1 on the combined dataset, along with the strongest subcategory (85.29%) and severity (89.11%) performance among all evaluated models. SHAP-based explainability analysis further validates the model's decision transparency and cross-lingual feature consistency, supporting its deployment in trustworthy, data-driven civic governance applications.

Keywords: Civic Complaint Classification, Hindi-English NLP, XLM-RoBERTa, Transformer Architecture, Multi-Task Learning, Code-Mixed Text, Smart Governance

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	Abstract	
	List of Tables	
	List of Figures	
	List of Abbreviations	
1	**INTRODUCTION**	1
1.1	Background of Civic Issue Reporting Systems	2
1.2	Problem Statement	2
1.3	Motivation	3
1.4	Objective	4
1.5	Scope of the Project	4
1.6	Existing System and Limitations	5
1.7	Proposed Solution	5
2	**LITERATURE REVIEW**	6
2.1	Civic Complaint Management	7
2.2	NLP and Complaint Classification	7
2.3	Multilingual and Code-Mixed NLP	8
2.4	Transformer-Based Approaches	8
2.5	Complaint Severity and Prioritization	8
2.6	Research Gap	9
2.7	Contribution of the Proposed Work	9

CHAPTER NO.	TITLE	PAGE NO.
3	**DATASET AND PROPOSED METHODOLOGY**	10
3.1	CivicComp Dataset	10
3.2	Dataset Construction	12
3.3	Synthetic Data Generation	12
3.4	Data Preprocessing	13
3.4.1	Data Cleaning	13
3.4.2	Text Normalization	13
3.4.3	Label Processing	13
3.4.4	Tokenization	14
3.4.5	Padding and Truncation	14
3.5	Dataset Structure and Labels	15
3.6	Proposed HIEN-BERT Architecture	15
4	**SYSTEM DESIGN AND ARCHITECTURE**	17
4.1	Overall System Architecture	17
4.2	Data Flow	18
4.3	Complaint Processing Workflow	19
4.4	Proposed Application Architecture	19
4.5	System Components	20
5	**METHODOLOGY**	21
5.1	Development Environment	21
5.2	Model Implementation	21

CHAPTER NO.	TITLE	PAGE NO.
5.3	Baseline Models	22
5.4	HiENBERT Implementation	22
5.5	Streamlit Prototype	23
5.6	Prototype Workflow	23
6	**RESULTS AND ANALYSIS**	25
6.1	Experimental Setup	25
6.2	Evaluation Metrics	25
6.3	Original Dataset Results	25
6.4	Synthetic Dataset Results	26
6.5	Combined Dataset Results	26
6.6	Language-Wise Analysis	26
6.7	Category-Wise Analysis	27
6.8	Confusion Matrix and t-SNE Analysis	28
6.9	SHAP Explainability	28
6.10	Discussion	29
7	**EXPERIMENTAL RESULTS**	30
7.1	System Testing	30
7.2	Current Achievements	30
7.3	Limitations	30
8	**CONCLUSION AND FUTURE WORK**	32
8.1	Conclusion	32

CHAPTER NO.	TITLE	PAGE NO.
8.2	Future Work	32
9	**REFERENCES**	33
10	**APPENDICES**	36
10.1	Sample Inputs and Outputs	36
10.2	Additional Graphs and Charts	37
10.3	Gantt Charts	37

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
2.1	Research Gap	9
3.1	CivicComp Dataset Summary	11
3.2	Hierarchical Dataset Structure	15
4.1	Major System Components	20
5.1	Development Environment	21
6.1	Original Dataset Performance	25
6.2	Synthetic Dataset Performance	26
6.3	Combined Dataset Performance	26
7.1	System Testing	30

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.1	Construction Pipeline for the CivicComp Dataset	12
3.2	Proposed HiENBERT Architecture	16
4.1	Overall System Architecture	18
5.1	Streamlit Prototype Interface	23
6.1	Language-Wise Performance	26
6.2	Category-Wise Performance	27
6.3	Confusion Matrix	28
6.4	t-SNE Visualization	28
6.5	SHAP Feature Importance	29
10.3	Design Diagram	37

LIST OF SYMBOLS AND ABBREVIATIONS

ABBREVIATION		EXPANSION
AI	-	Artificial Intelligence
NLP	-	Natural Language Processing
LLM	-	Large Language Model
ML	-	Machine Learning
DL	-	Deep Learning
BERT	-	Bidirectional Encoder Representations from Transformers
mBERT	-	Multilingual BERT
XLM-R	-	Cross-lingual RoBERTa
IndicBERT	-	Multilingual BERT for Indian Languages
SVM	-	Support Vector Machine
TF-IDF	-	Term Frequency – Inverse Document Frequency
RNN	-	Recurrent Neural Network
LSTM	-	Long Short-Term Memory
BiLSTM	-	Bidirectional Long Short-Term Memory
API	-	Application Programming Interface
GPU	-	Graphics Processing Unit
CSV	-	Comma-Separated Values
JSON	-	JavaScript Object Notation
BBMP	-	Bruhat Bengaluru Mahanagara Palike
PII	-	Personally Identifiable Information
BLEU	-	Bilingual Evaluation Understudy
IoU	-	Intersection over Union
TP	-	True Positive
TN	-	True Negative
FP	-	False Positive
FN	-	False Negative

CHAPTER 1
INTRODUCTION

The process of urbanization and the high rate of population growth has heightened the need to have efficient civic infrastructure as well as service management. Among the problems that cities are challenged with are waste disposal, road network issues, water supply concerns and safety of the population. These problems are addressed under the grievance redressal systems which are more or less manual and they do not have intelligent mechanisms to process high number of complaints. It leads to time wastage, wrong classification and inefficient routing; which in the end lowers the quality of services as well as the satisfaction of the citizens.

The development of Artificial Intelligence (AI) and Natural Language Processing (NLP) allows changing the conventional system of complaints into an automated, intelligent system. These technologies can handle unstructured text information, derive insights and aid decision-making. Transformer-based models have demonstrated good use in text classification and multilingual comprehension thus applicable in civic complaint analysis.

The proposed project is the automated system of civic issues reporting, classification, and prioritization based on AI. The system is designed to make the system more efficient, lower response time, and increase transparency by using NLP and transformer models. This helps towards attaining the vision of smart governance and citizen-centric service delivery.

1.1 Background of Civic Issue Reporting Systems

The systems of urban civic issue reporting are crucial to sustaining urban infrastructure and civic services. In the past, they used to take up complaints by visiting them, calling them on the phone or putting down an application. Such means were complicated, lacked tracking systems and were not efficient.

Governments have become more accessible and transparent with online grievance portals and mobile apps, which have been brought about by digitalization. Nevertheless, these systems continue to rely on a manual method of classification, prioritization, and routing.

One of the crucial shortcomings is unstructured and multilingual data. Complaints provided by citizens are usually in other languages or in other informal formats making it hard that the traditional systems decipher them correctly. Moreover, there are no automated prioritization and this contributes to the delay in solving urgent problems.

AI and NLP enter the civic systems allow the automated processing of the texts and their classification as well as the support of the decisions. This is needed to enhance efficiency, scalability and responsiveness in urban governance.

1.2 Problem Statement

City governments are bombarded with complaints of civic nature like garbage disposal, road repair, water supply and safety matters. Even with the digital platforms, there are a number of challenges.

The major problem is the manual classification and routing of complaints since unstructured inputs cannot be interpreted easily causing misclassification and delays. The other challenge is the multilingual data processing particularly where complaints are made in the local languages and the code-mixed languages such as the Hinglish language in India. The available systems also do not contain sufficient language processing abilities that restrict accessibility.

Moreover, there is no proper prioritization system and thus urgent issues are postponed. The existing systems also have no intelligent analysis, including semantic understanding and pattern recognition, so the resources allotment and coordination are not efficient.

Thus, it is necessary to have an automated, scalable system to correctly classify, prioritize and route complaints and process multilingual and unstructured data in an efficient manner.

1.3 Motivation

The rising civic complaints as a result of urban development is a testimony to the need to have a more effective system. The current systems are inefficient and slow due to the need to use manual systems that create delays and dissatisfaction among the citizens.

The main reason is to enhance efficiency and responsiveness through automation of classification

and routing of complaints. This saves human labor, errors are minimized and speed is increased. The next major driving force is inclusiveness, where the multilingual support enables the user to leave complaints in the language of choice.

Prioritization is also required to be smart since not every issue should be urgent. The classification of the issues based on severity guarantees that important issues are tackled within the shortest time possible. Moreover, the development of the models based on transformers gives the chance to create the correct systems that can comprehend the complicated language patterns.

The aim of smart governance which is to increase transparency, accountability and the capability of making data-based decisions as well as the enhancement of citizen engagement further motivates the project.

1.4 Objective

The main objective of the proposed project is to develop an intelligent multi-lingual civic complaint processing framework that can automatically understand, classify and prioritize citizen complaints submitted in English, Hindi and Hinglish. The project aims to build and utilize the CivicComp dataset consisting of 60,708 multilingual complaint samples annotated with category, subcategory and severity, and to develop HiEN-BERT, a transformer-based framework for dealing with multilingual and code-mixed civic complaints. The system aims to assist the efficient grievance management through automated single-label and multi-label classification of complaints, severity prediction and priority assignment. The purpose is also to compare different machine-learning and transformer-based models using suitable evaluation metrics and integrate the trained models into a functional web application prototype that demonstrates the complete complaint processing workflow while reducing manual effort and improving the efficiency of civic grievance handling.

1.5 Scope of the Project

The project is aimed at creating an AI-driven apparatus that will be used to analyze, categorize, and prioritize civic complaints automatically. It involves the architecture of an entire pipeline, including data collection, preprocessing, model training and evaluation.

The system manages multilingual information such as English, Hindi and Hinglish and category, subcategory and severity have been annotated. Transformer based models which include IndicBERT and XLM-RoBERTa are employed to achieve proper classification and contextual comprehension.

There is also the automatic routes of complaints to the respective departments and prioritization based on the severity to enhance efficiency. A comparative analysis of the various models is carried out to determine the best model.

It is confined in text analysis and excludes image and video processing, complete user interface development, and real-time implementation in government systems.

1.6 Existing System and Limitations

Current systems of civic grievances have a number of shortcomings. They are also very dependent on manual processing that is time consuming and is subject to errors. Such systems are also not good in unstructured and informal text, which results in erroneous interpretation.

The support is multilingual which minimizes the availability of the services to users who use regional or mixed languages. Also, the prioritization mechanism is not effective, and urgent problems take time to resolve.

The next weakness is the absence of smart analytics as the existing systems are not able to recognize trends or offer decision support. Poor route arrangement also contributes to time wastage and lack of adequate coordination among the departments.

These restrictions demonstrate the necessity of a smart and automated system to enhance the accuracy, efficiency and user experience.

1.7 Proposed Solution

The proposed solution is an AI-based multilingual civic complaint processing and prioritization framework to perform the automated initial analysis of citizen grievances in English, Hindi and Hinglish. The system receives a complaint through the application interface, performs necessary text preprocessing and processes the complaint using the developed multilingual NLP models. The

proposed framework is based on HiEN-BERT which is used to identify the relevant civic issue category and subcategory and the system also determines the severity of the complaint and supports priority assignment. The framework consists of single-label and multi-label classification to address complaints with one or more civic issues and integrates the trained models into a working web application prototype. The complete workflow is Complaint Submission, Text Pre processing, Multilingual processing, Category/Subcategory Classification, Severity Prediction, Priority Assignment. This provides a structured and automated way to help civic authorities manage and prioritize incoming complaints efficiently.

CHAPTER-2

LITERATURE REVIEW

Artificial Intelligence (AI) and Natural Language Processing (NLP) have evolved, which has greatly enhanced the complaint management systems. Previously, there were manual systems/rule-based systems, which were not scalable, and could not work with unstructured data. Some works by Fauzi presented ensemble classifiers to classify complaints, and Rao et al. revealed the utilization of NLP in voice-based complaints.

Recent progress lays stress on transformer-based and on deep learning methods. Esperança et al. used BERT embeddings to classify semantically achieving higher accuracy and less redundancy. In another study, Sierra et al. and Rakhimzhanov et al. also indicated the usefulness of Large Language Models (LLMs) in the automated classification of complaints with high efficiency.

There has also been multimodal and multi-task learning methods. Singh et al. presented classification with the severity detection, whereas another study by Singh et al. included the text and image data in order to improve the analysis. Fang and Shan and Gilman et al. highlighted the user-centric systems and data issues like scalability and privacy in the smart city setting.

Transformer-based models have been effective in languages with multilingual settings. Agbesi et al. and Yadav et al. emphasized their performance in both low-resource and code-mixed languages, whereas Salleh et al. suggested hybrid methods that can improve multilingual processing. The systems created by Shama et al., Padmane et al., and Dash et al. using applications prove the necessity of dealing with various input formats.

Graph-based and federated learning are also some of the advanced learning techniques examined. Graph contrastive learning was proposed by Zhang et al., and federated multi-task learning was proposed by Singh et al. Wang et al. had to deal with privacy issues in such systems.

2.1 Civic Complaint Management

The processing of civic complaints has evolved from rule-based and manual processes to AI-assisted systems to deal with large volumes of grievances from citizens. The first approaches were mostly based on manual categorization and traditional machine learning approaches while the more recent systems are based on NLP, deep learning and large language models for automated complaint processing. [19] Fauzi looked into automatic complaint classification with an ensemble of classifiers. Esperança et al. [1], Sierra et al. [2] and Rakhimzhanov et al. [3] showed the use of AI, LLMs and language models for complaint management and classification. The studies indicate the potential of automation but most of the existing systems are domain specific and do not provide integrated multilingual classification, severity analysis and prioritization**.**

2.2 NLP and Complaint Classification

Natural Language Processing (NLP) allows for the automatic analysis of free-form and unstructured civic complaints. Early work utilized statistical features, keywords, and conventional machine-learning classifiers, with Fauzi [19] investigating classifier ensembles for complaint classification. More recent work has employed semantic and contextual representations, such as the work of Esperança et al. [1] and Rakhimzhanov et al. [3]. Recent approaches include transformer-based and large language model approaches [2], graph-based and hybrid approaches [15], [16], and multi-task learning [12]. However, there are challenges in dealing with multilingual, code-mixed, domain-specific and multi-label civic complaints.

2.3 Multilingual and Code-Mixed NLP

Multilingual NLP is crucial for civic systems, as citizens may express their complaints in different languages or mix languages within the same sentence. [8] Agbesi et al. explored transformer based classification for low resource languages. [10] Yadav et al. focused on code-mixed language processing. Mishra et al. [25] studied the domain adaptive BERT for Hinglish text and showed the importance of contextual models for code-mixed Indian languages. Padmane et al. [20] have also worked on multilingual speech and text processing. While these studies do support multilingual and code-mixed processing, not much work has been done specifically on English, Hindi and Hinglish civic complaints within a unified complaint-management framework.

2.4 Transformer-Based Approaches

Transformer models have pushed the state of the art in NLP with self-attention capturing contextual relationships in text. BERT based and other transformer architectures have been used for complaint classification, low resource languages and code-mixed text [1], [8], [25]. LLMs have been also studied for complaint management and classification in recent studies [2], [3]. These methods have superior context awareness compared to many classical approaches but their performance relies on appropriate domain-specific data and computational resources. Thus, tailoring transformer models for multilingual civic complaints remains an important research topic.

2.5 Complaint Severity and Prioritization

Complaint classification by itself does not indicate how urgent a complaint is. Severity detection and prioritization can help identify critical complaints and help in effective resource allocation. Singh et al. [6] worked on complaint and severity identification and Singh et al. [7] worked on multimodal complaint severity detection. Lidiawaty et al. [13] proposed an urgency-based approach to prioritize complaint texts, and recent work has investigated LLM-assisted severity classification and complaint triaging [23], [24]. However, these approaches are mostly domain-specific, which entails the integration of severity and prioritization mechanisms for multilingual civic complaints.

2.6 Research Gap

The studies presented here show considerable advances in AI-based complaint handling, NLP classification, multilingual processing, transformer models, and severity detection. But those capabilities are often developed in isolation, or for specific domains. There are very few studies that have combined English, Hindi, and Hinglish processing with civic complaint classification, multi-label analysis, severity prediction, and prioritization. There is also a lack of large scale domain specific datasets with structured multilingual civic complaints with category, subcategory and severity annotations. These limitations motivate us to develop an integrated framework for multilingual civic complaint processing .

Table 2.1 Research Gap

Research Area	Existing Limitation	Proposed Direction
Multilingual processing	Limited civic-domain Hindi/Hinglish support	English, Hindi and Hinglish
Complaint classification	Mainly single-task approaches	Category and subcategory classification
Multi-label analysis	Limited handling of multiple issues	Multi-label complaint classification
Dataset	Limited multilingual civic datasets	CivicComp dataset
Severity	Often domain-specific	Integrated severity prediction
Prioritization	Limited integration with classification	Automated priority assignment
Application	Separate research components	End-to-end framework

2.7 Contribution of the Proposed Work

The proposed work addresses these gaps with an integrated AI-based multilingual civic complaint processing framework using the CivicComp dataset consisting of 60,708 English, Hindi and Hinglish complaint samples. The framework utilizes HiEN-BERT for contextual complaint understanding and provides complaint category, subcategory, multi-label classification, severity prediction and prioritization. Different machine-learning and transformer-based methods are evaluated, and the trained models are integrated into a working web application prototype. Therefore, the proposed work integrates multilingual NLP, complaint classification, severity analysis and prioritization into a single system for intelligent civic grievance management.

CHAPTER 3

DATASET AND PROPOSED METHODOLOGY

3.1 CivicComp Dataset

An effective automated civic complaint classification system requires a dataset that captures the language, diversity, and hierarchical nature of real-world civic complaints. To address this need, CivicComp, a multilingual civic complaint dataset, was created for this project. The dataset consists of complaints in English, Hindi and Hinglish, so the proposed system can handle both standard multilingual and code-mixed complaints.

CivicComp's design is based on a hierarchy of complaint types: Primary Category, Subcategory, and Severity. The main category is the general type of civic issue and the sub-

category is the specific problem. Severity is the urgency of the complaint. For instance, a complaint regarding the availability of the water can be classified as Water Supply at the first level and No Supply or Poor Supply at the subcategory level. A severity label is then applied to communicate the urgency of the issue.

The final dataset consists of 60,708 multilingual samples across 8 primary categories, 24 subcategories and 3 severity levels. The dataset contains original data of complaints and synthetic generated complaints. Of these, 15,000 complaints were synthetically generated using multiple Large Language Models (LLMs) to enhance data diversity and representation.

Table 3.1 CivicComp Dataset Summary

Attribute	**Description**
Dataset	CivicComp
Languages	English, Hindi, Hinglish
Final Samples	60,708
Primary Categories	8
Subcategories	24
Severity Levels	3
Synthetic Complaints	15,000
Prediction Tasks	Category, Subcategory, Severity

The dataset provides the foundation for training, comparing, and evaluating multilingual transformer models for automated civic complaint analysis.

3.2 Dataset Construction

The CivicComp dataset was generated through a multi-step process of complaint collection, data preparation, synthetic data generation, multilingual processing, label organization and dataset refinement.

The general construction process is as follows:

Complaint Collection → Data Cleaning → Category Mapping → Synthetic Data Generation → Multilingual Processing → Dataset Refinement → Final CivicComp Dataset

The original complaint data contains examples of real civic issues. Synthetic complaints are added to increase the availability of samples for different complaint categories and linguistic variations.

Collected and generated complaints were processed to retain a consistent hierarchical labeling structure. We screened the samples for inappropriate or duplicate content and mapped the category information to the predefined CivicComp categories.

The final dataset was then ready for model training and evaluation.

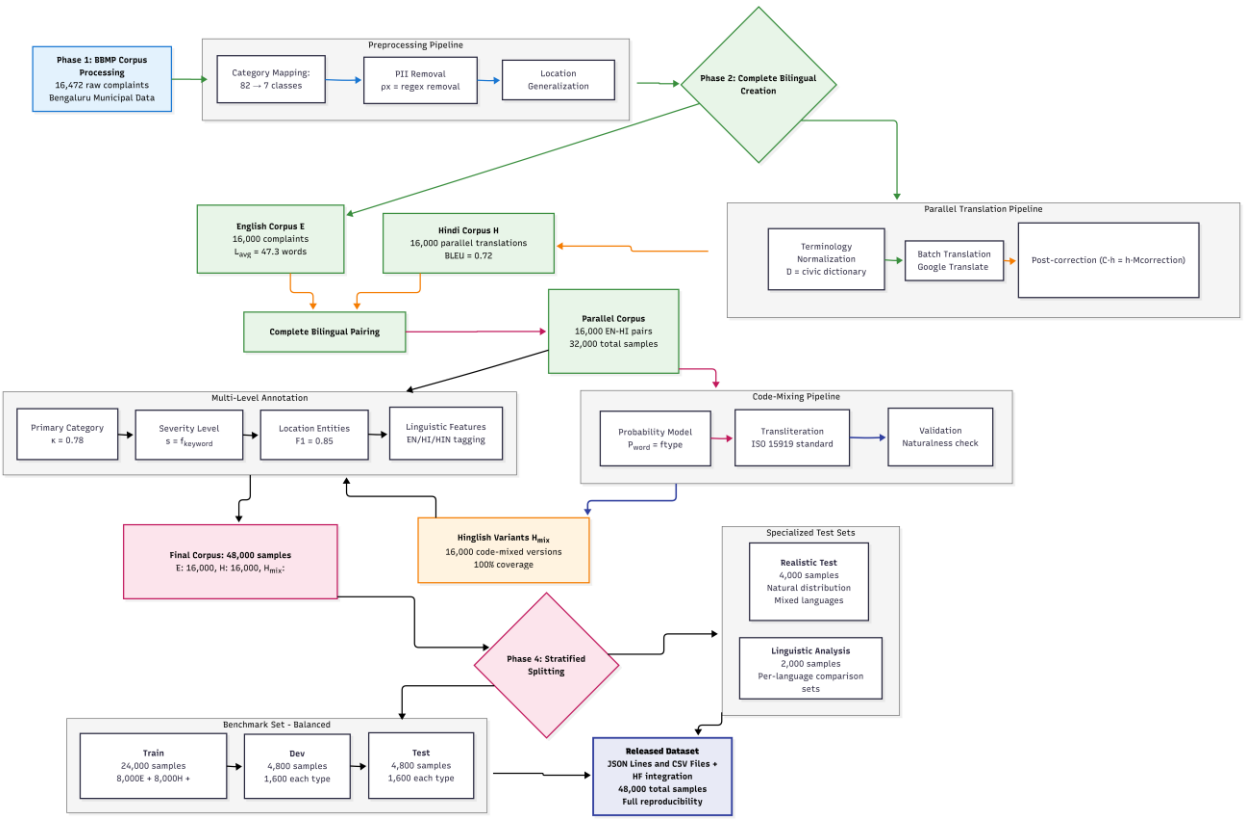


Figure 3.1: Construction pipeline for the CivicComp Dataset

The construction process guaranties that the final dataset has sufficient linguistic variation, while maintaining the same complaint categories and prediction labels.

3.3 Synthetic Data Generation

Another important aspect of CivicComp is the use of synthetic complaint generation to increase the size and diversity of the training data. In total, five Large Language Model sources were used to generate 15,000 synthetic civic complaints:

- ChatGPT
- Claude
- Gemini
- DeepSeek
- Qwen

The use of multiple LLMs helps to generate complaints with different writing patterns and expressions. This is especially useful for civic complaints, since different citizens may use different vocabularies, sentence structures, detail levels, and language combinations to describe the same problem. Generated complaints were mapped to the predefined civic complaint categories and processed as part of the dataset refinement pipeline. Samples that did not meet the required characteristics of the data were removed or excluded during refinement. Synthetic data was used as a data augmentation mechanism, and not as a replacement of the original complaint data. The final CivicComp dataset is constructed by merging available original and synthetic complaint data to build a larger multilingual dataset for model development.

The synthetic data generation step is especially important to enhance the representation of the less frequent complaint categories and to provide additional linguistic variations for the model training.

3.4 Data Preprocessing

Before training the classification models, the complaint data was preprocessed to improve the consistency and compatibility with the transformer-based models.

The main preprocessing operations were:

3.4.1 Data Cleaning

Incomplete, unusable or inappropriate complaint records were identified and deleted. Dataset refinement also included handling of duplicate or redundant samples.

3.4.2 Text Normalization

The complaint text was normalized so as to reduce unnecessary variations, but keeping the meaningful words and expressions that are necessary for the classification. The preprocessing was done to retain the important linguistic information for Hindi, English and Hinglish complaints.

3.4.3 Label Processing

The main category, subcategory and severity labels were transformed into numerical representations that can be used for training the model.

3.4.4 Tokenization

The cleaned complaint text was converted into tokens that were compatible with the model using the tokenizer of the selected transformer model. Tokenization essentially transforms natural language into numerical input representations that can be fed into the transformer encoder.

3.4.5 Padding and Truncation

As complaints can be of different lengths, the tokenized sequences were either padded or truncated to a desired maximum sequence length. This allows multiple complaints to be processed in a batch during training.

The preprocessing pipeline can thus be summarized as:

Raw Complaint → Cleaning → Normalization → Label Encoding → Tokenization → Padding/Truncation → Model Input

The same general pre-processing procedure was kept across the evaluated transformer models to ensure a consistent experimental set-up.

3.5 Dataset Structure and Labels

CivicComp uses a three-level hierarchical label structure with three levels of prediction:

Primary Category → Subcategory → Severity

The first category is the generic civic issue the citizen reported. The dataset contains 8 major categories of types of civic complaints.

Subcategory:

For example:

Water Supply → No Supply or Water Supply → Poor Supply

Severity

The subcategory describes the issue more specifically in the respective primary category. There are 24 subcategories in the dataset.

Each complaint is also given one of three levels of severity. Severity is an indication of the importance of the reported issue and is modeled as a separate prediction task in the proposed model.

Table 3.2 Hierarchical Dataset Structure

Level	**Purpose**	**Example**
Primary Category	Identifies broad civic issue	Water Supply
Subcategory	Identifies specific issue	No Supply
Severity	Identifies urgency	Severity Level

The hierarchical structure enables the proposed system to provide more detailed information than a conventional single-label complaint classifier. It also provides for future automated complaint prioritization and routing.

3.6 Proposed HIEN-BERT architecture

We introduce a novel multilingual transformer-based framework HiENBERT for hierarchical classification of civic complaints and severity prediction.

The main idea is to utilize a transformer encoder to learn a contextual representation of the complaint, and then utilize the task-specific prediction heads to perform the three classification tasks.

The overall architecture is depicted as follows:

Input Complaint → Transformer Encoder → Shared Representation → Task-Specific Heads → Category / Subcategory / Severity

The two variants of the proposed framework are considered.

- HiENBERT-Indic, based on IndicBERT.
- HiENBERT-XLMR, based on XLM-RoBERTa.

The proposed framework can be evaluated on different multilingual representation capabilities using different transformer backbones.

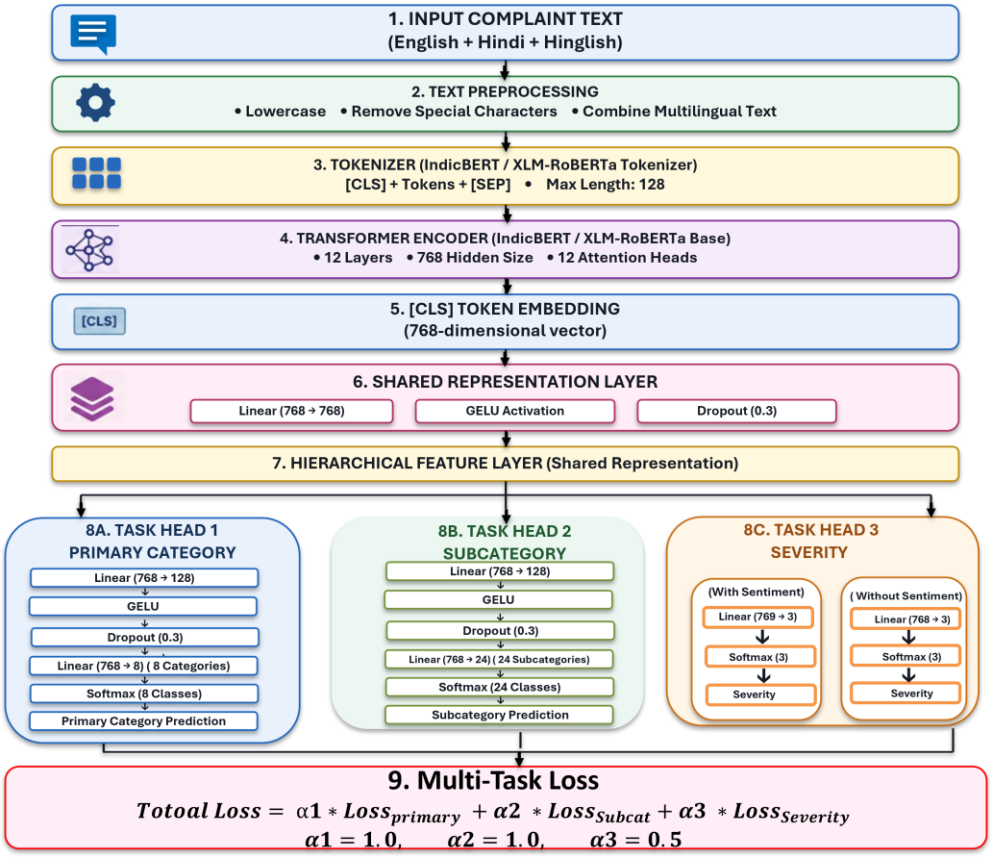


Figure 3.2: Proposed HiENBERT Architecture

The architecture consists of a shared representation layer and separate prediction heads. This shared representation allows information learned from the complaint text to be used for all three prediction tasks.

The three output heads are:

1. Primary Category Head
2. Subcategory Head
3. Severity Head

This design allows a single model to handle multiple related classification tasks simultaneous ly.

CHAPTER 4

SYSTEM DESIGN AND ARCHITECTURE

4.1 Overall System Architecture

The proposed system is supposed to be a multilingual AI-based civic complaint processing system that takes in citizens' complaints in text and automatically analyzes them using the trained HiENBERT model. The system is designed to process complaints in English, Hindi and Hinglish and give out structured information to help in complaint management.

The whole system is composed of four major phases:

Complaint Input → Text Processing → HiENBERT Prediction → Structured Output

The complaint given by user is processed and converted to a format that is compatible with the trained transformer model. HiENBERT then analyzes the complaint and predicts the main category, subcategory and severity. The prediction results are displayed in the application interface.

The main elements of the system are:

1. User Interface – receives the civic complaint from the user.
2. Input Processing Module – performs the required pre-processing and tokenization.
3. HiENBERT Model – is a multi-lingual complaint classification model.
4. Prediction Module - predicts category, subcategory and severity.
5. Explainability Module - model interpretation using SHAP.
6. Application Layer - displays the prediction results to the user.

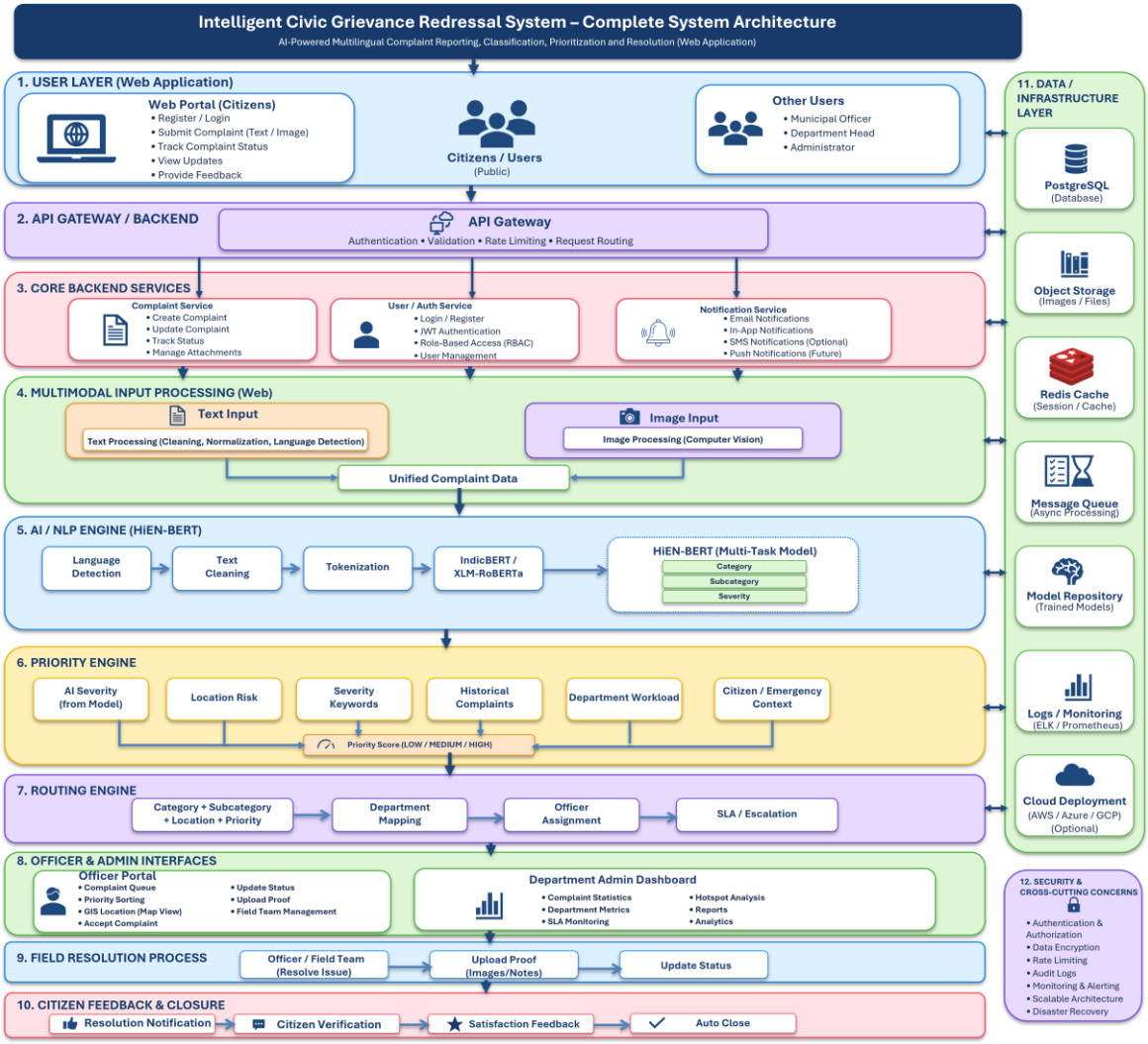


Figure 4.1 Overall System Architecture

This current version includes core AI functionality based in a Streamlit prototype, while the full web application and backend integration are part of the ongoing development.

4.2 Data Flow

For a complaint, the system operates through the following sequence:

Complaint Input → Tokenization → Transformer Encoder → Shared Representation → Prediction Heads → Output

The tokenizer turns the complaint into inputs the model can understand. The HiENBERT encoder generates contextual representations, which are fed to three prediction heads for category, subcategory and severity. The results are then shown in the interface of the application.

4.3 Complaint Processing Workflow

The complaint processing workflow includes the following steps:

1. Complaint Submission – User submits civic complaint.
2. Preprocessing – Cleaned and tokenized text.
3. Model Inference – HiENBERT takes the complaint.
4. classification - Category, subcategory, and severity.
5. Result Display – Display predictions and confidence information.
6. Explainability – SHAP can be used to analyze important textual features.

The structured information obtained can help in future complaint prioritization and routing.

4.4 Proposed Application Architecture

The application is divided into three main layers:

Presentation Layer : Offers the interface to lodge complaints and view predictions. The present prototype is implemented using Streamlit.

AI/Applicaton Layer: Pre-processes, infers model, generates predictions and explainability with trained HiENBERT model

Data Layer / Backend: The web application will provide complaint storage, management and integration with backend services. This component is being developed.

4.5 System Components

Table 4.1 Major System Components

Component	**Function**
User Interface	Accepts complaints and displays results
Preprocessing Module	Prepares complaint text
Transformer Model	Generates contextual representations
Category Classifier	Predicts primary category
Subcategory Classifier	Predicts specific issue
Severity Classifier	Predicts severity

Explainability Module	Provides SHAP-based analysis
Backend/Database	Planned complaint management and storage

The components work in concert to convert an unstructured multilingual complaint into structured classification results.

CHAPTER-5
METHODOLOGY

5.1 Development Environment

The project was implemented using Python and deep learning libraries for dataset processing, model training and evaluation, as well as prototype development.

Table 5.1 Development Environment

Component	**Technology**
Programming Language	Python
Deep Learning	PyTorch
NLP Framework	Hugging Face Transformers
Data Processing	Pandas, NumPy
Model Evaluation	Scikit-learn
Explainability	SHAP
Prototype	Streamlit
Development	Jupyter Notebook / Google Colab
Hardware	GPU-enabled environment

5.2 Model Implementation

Trained and evaluated transformer models were based on the prepared CivicComp data set.
The complaint text was tokenized and fed into the respective transformer backbone.
The implementation consists of:

- Dataset loading and preprocessing
- Tokenization
- Model training
- Validation and evaluation
- Prediction generation
- Performance comparison

The complaint classification tasks are predicted by the trained models.

5.3 Baseline Models

Several multilingual transformer models were implemented as baselines to compare with the proposed approach . The models tested are as follows:

- IndicBERT
- XLM-RoBERTa
- mBERT
- MuRIL

For a fair comparison, the same dataset and evaluation method are used to test these models.

5.4 HiENBERT Implementation

We implemented HiENBERT as a multi-
task classification model using transformer backbones.
The implementation uses a common representation and separate prediction heads for:

- Primary Category
- Sub-category
- Severity

Two different versions were tried:

HiENBERT-Indic with IndicBERT and HiENBERT-XLMR with XLM-RoBERTa.
The trained model is saved and loaded during inference so that new complaints can be classifi ed via the prototype.

5.5 Streamlit Prototype

A prototype of the main functionality of the proposed system was implemented on Streamlit.

The prototype lets the user:

1. Write a civic report.
2. Run the trained model on the complaint.
3. Predict category, subcategory and severity. 4.
4. Show the prediction results and confidence information.
5. View explainability information where applicable.

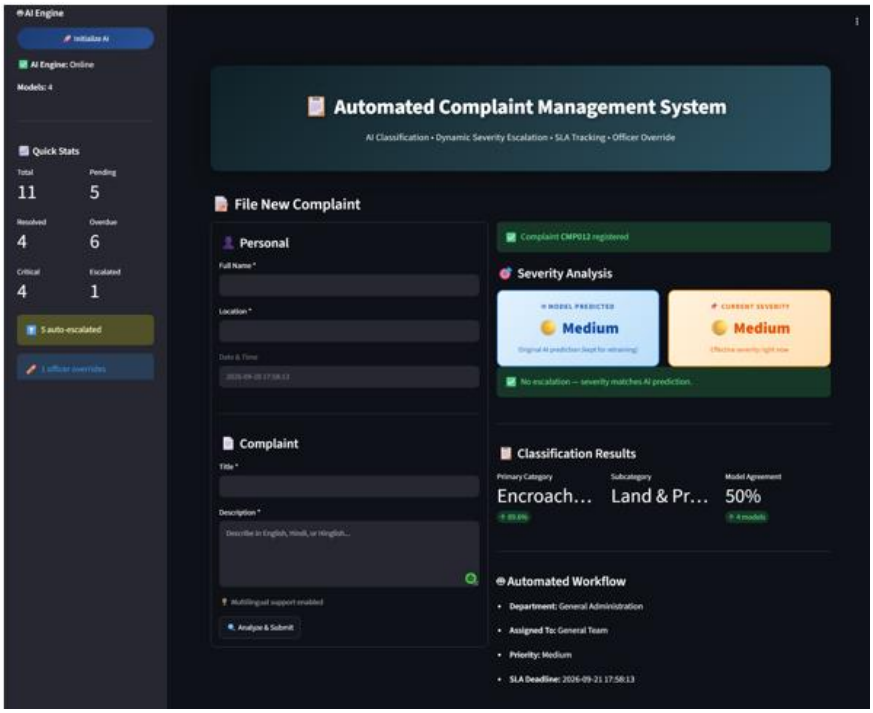


Figure 5.1: Streamlit Prototype Interface

The prototype shows that the trained AI model can be integrated into an interactive application and is not restricted to offline experiments.

5.6 Prototype Workflow

The prototype has a simple workflow:

User Complaint –> Text Processing –> Trained HiENBERT –> Prediction –> Result Display

The user inputs a complaint through the interface. The application pre-processes the input and passes it to the trained model. HiENBERT produces the required predictions and then displays them via the Streamlit interface.

The current prototype shows the basic AI functions. The full web application is in development with backend integration and additional complaint management features.

CHAPTER -6

RESULTS AND ANALYSIS

6.1 Experimental Setup

The trained baseline models and HiENBERT variants were evaluated on the CivicComp dataset with the same train-validation-test setup as above. We experimented on the original, synthetic and combined datasets to see the effect of data augmentation and multilingual training.

The models evaluated are IndicBERT, XLM-RoBERTa, mBERT, MuRIL, HiENBERT-Indic and HiENBERT-XLMR.

6.2 Evaluation Metrics

Standard classification metrics were used to evaluate the models:

- Accuracy – the fraction of all predictions that are correct.
- Precison – percentage of positive classes predicted correctly.
- Recall – % of true classes recognized correctly.
- Macro-F1 – mean F1-score of the classes, treating all classes equally.
- Weighted-F1 - F1 score weighted by class frequency.

These measures give a balanced evaluation especially for datasets with classes of different frequencies.

6.3 Original Dataset Results

Model performance was assessed on the original dataset prior to synthetic data augmentation.

The results demonstrate the capability of the transformer models to classify the civic complaints based on the original data available. We found differences between models for the category, subcategory and severity tasks.

Table 6.1 Original Dataset Performance

Model	Accuracy	F1-Macro	F1-Weighted	Precision	Recall	Cohen's Kappa	MCC
IndicBERT	0.8644	0.758	0.8608	0.8613	0.864	0.8095	0.81
XLM-RoBERTa	0.8708	0.7735	0.8709	0.8722	0.871	0.8216	0.822
MuRIL	0.8734	0.7598	0.8716	0.8726	0.873	0.8239	0.824
M-BERT	0.8689	0.759	0.8664	0.8664	0.869	0.8165	0.817
HiENBERT-Indic	0.8695	0.7241	0.8655	0.8651	0.87	0.8178	0.818
HiENBERT-XLMR	0.8663	0.7276	0.8646	0.8638	0.866	0.8145	0.815

The results are used as a baseline to compare with the synthetic and combined datasets.

6.4 Synthetic Dataset Results

We also evaluated the synthetic dataset separately to investigate the impact of generated complaint data on classification performance.

Table 6.2 Synthetic Dataset Performance

Model	Accuracy	F1-Macro	F1-Weighted	Precision	Recall	Cohen's Kappa	MCC
IndicBERT	0.948	0.9476	0.9477	0.948	0.948	0.9406	0.941
XLM-RoBERTa	0.9631	0.9631	0.9631	0.9635	0.963	0.9578	0.958
MuRIL	0.9587	0.9585	0.9585	0.959	0.959	0.9528	0.953
M-BERT	0.964	0.964	0.964	0.9644	0.964	0.9589	0.959
HiENBERT-Indic	0.9436	0.9435	0.9435	0.9466	0.944	0.9355	0.936
HiENBERT-XLMR	0.9631	0.963	0.963	0.9631	0.963	0.9578	0.958

The results show the influence of additional generated samples to model learning and provide a basis for evaluating the utility of synthetic data augmentation.

6.5 Combined Dataset Results

The combined dataset contains the cleaned multilingual data used for the main evaluation of the proposed framework.

On the combined dataset, HiENBERT-XLMR got 93.28% accuracy and 93.42% Macro-F1. It achieved 85.29% Macro-F1 for subcategory classification and 89.11% for severity classification.

Table 6.3 Combined Dataset Performance

Model	Accuracy	F1-Macro	F1-Weighted	Precision	Recall	Cohen's Kappa	MCC
IndicBERT	0.9144	0.9147	0.9139	0.9152	0.914	0.901	0.901
XLM-RoBERTa	0.9321	0.9334	0.9321	0.933	0.932	0.9216	0.922
MuRIL	0.9351	0.9365	0.935	0.9355	0.935	0.9251	0.925
M-BERT	0.9308	0.9325	0.9308	0.9315	0.931	0.9201	0.92
HiENBERT-Indic	0.9167	0.9166	0.9164	0.9171	0.917	0.9037	0.904
HiENBERT-XLMR	0.9328	0.9342	0.9328	0.9332	0.933	0.9224	0.923

These results show the effectiveness of the proposed multi-task framework for multilingual civic complaint classification.

6.6 Language-Wise Analysis

The performance over English, Hindi and Hinglish complaints were also studied. This analysis helps to assess if the models behave consistently over different language forms and code-mixed input.

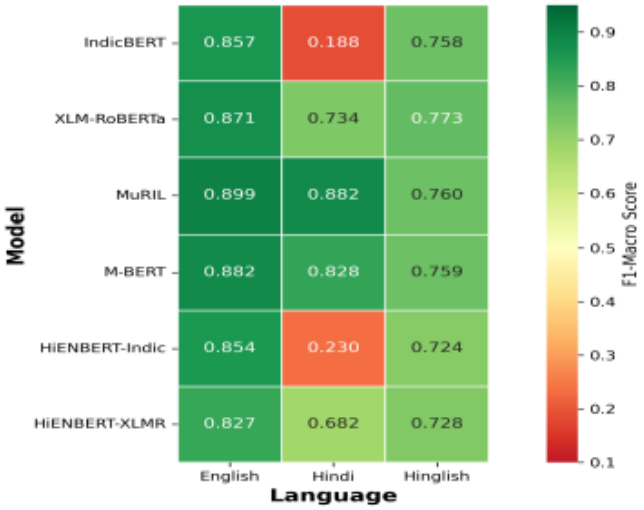


Figure 6.1 Language-Wise Performance

The language-wise results help identify the differences in model performance across multilingual complaint types.

6.7 Category-Wise Analysis

The analysis was performed based on the category to assess the model performance across different civic issue categories.

The analysis also assists in identifying categories that are more consistently classified and categories where there is confusion between related types of complaints.

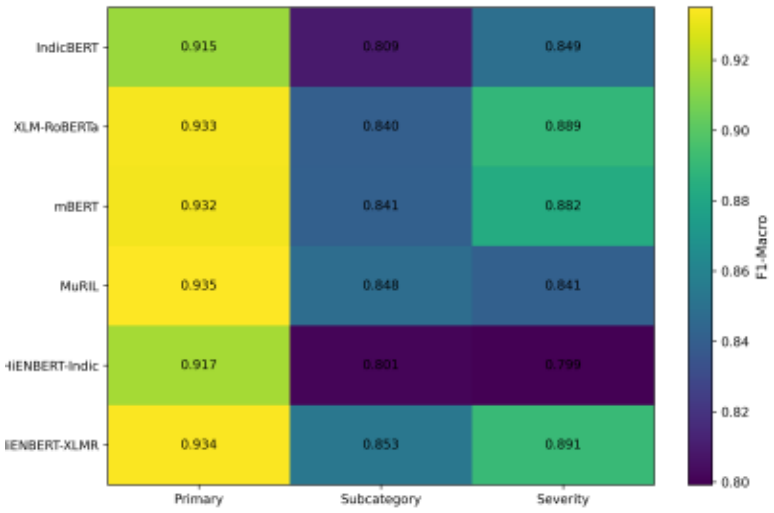


Figure 6.2 Category wise Performance

6.8 Confusion Matrix and t-SNE Analysis

Confusion matrix was used to find the incorrect prediction between the classes. It

provides a graphical representation of correctly and wrongly classified samples.

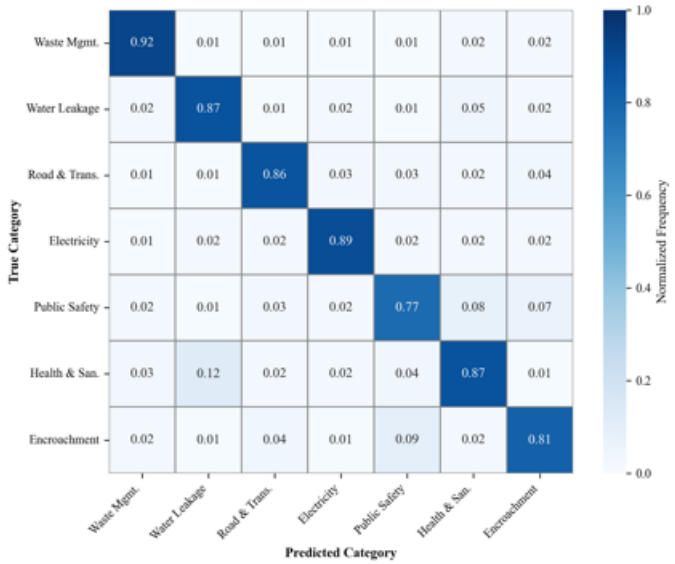


Figure 6.3: Confusion Matrix

t-SNE was used to visualize the learned feature representations in a two-dimensional space. It is useful to examine the degree of separation between different complaint classes.

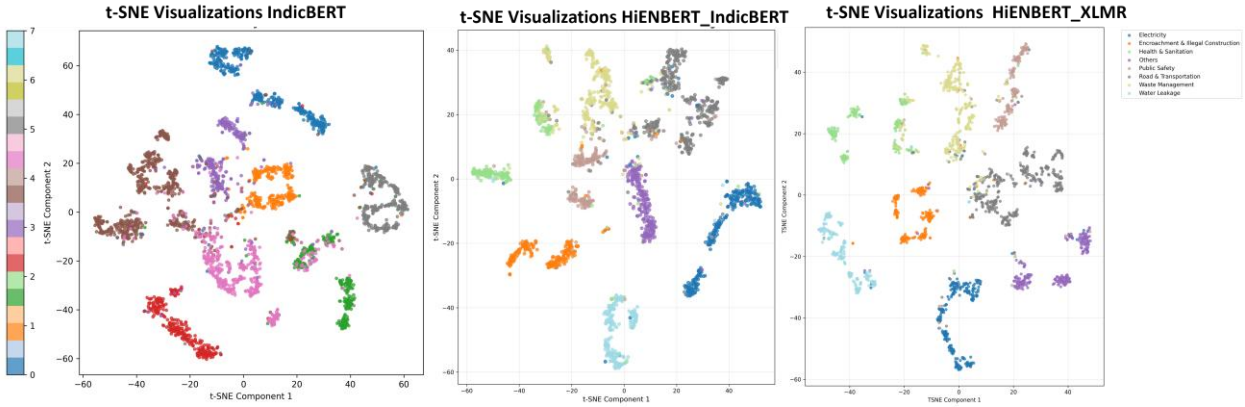


Figure 6.4: t-SNE Visualization

These analyzes in combination provide further insight into the classification behavior of the model beyond numerical performance indicators.

6.9 SHAP Explainability

SHAP (SHapley Additive exPlanations) was applied to study the contribution of input words or features to the model predictions.

The analysis helps in understanding which textual features impact the classification of

civic complaints and adds more transparency to the decision-making process of the model.

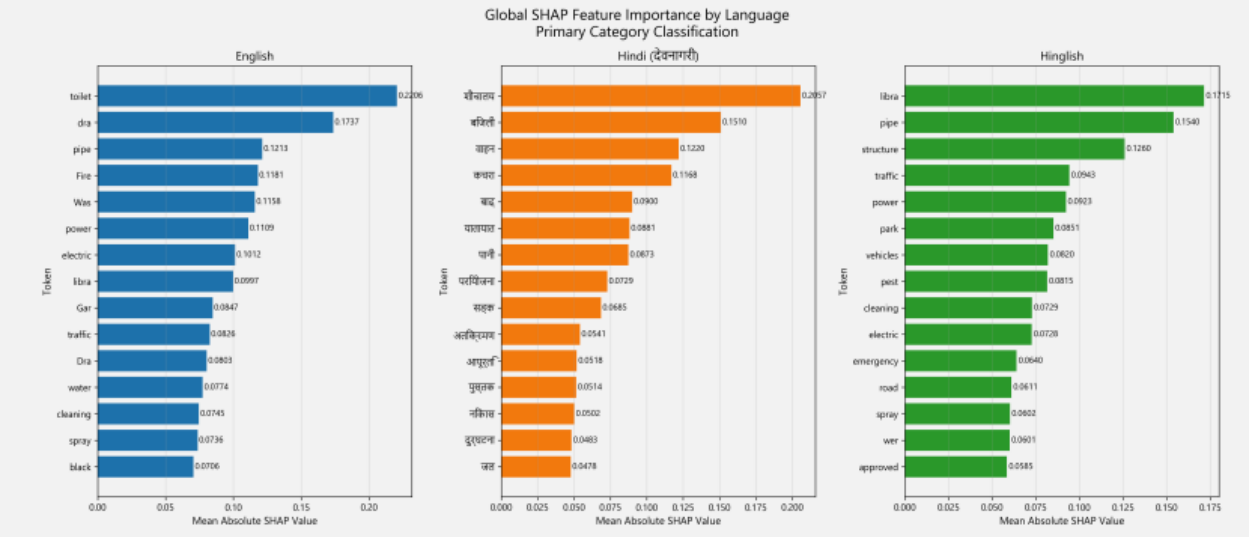


Figure 6.5 SHAP feature importance .

SHAP analysis provides interpretation of individual model predictions beyond the quantitative evaluation.

6.10 Discussion

The experiments show that transformer-based models can successfully process civic complaints in multiple languages. The combined dataset has a larger scope for training and evaluation . The proposed HiENBERT framework simultaneously addresses category , subcategory and severity prediction.

HiENBERT-XLMR obtained the best results on the combined dataset among all the tested configurations, thus proving the effectiveness of multilingual contextual representations and multi-task learning for the proposed application.

Results, confusion matrix, t-SNE visualization and SHAP analysis provide a quantitative and qualitative evaluation of the proposed system.

CHAPTER -7
EXPERIMENTAL RESULTS

7.1 System Testing

The developed prototype was tested to check the basic functionality of the civic complaint classification system. Test cases on complaint input, multi language processing, model prediction and result display

Table 7.1 System Testing

Test Case	Expected Result	Status
English complaint input	Complaint is processed correctly	Passed
Hindi complaint input	Complaint is processed correctly	Passed
Hinglish complaint input	Complaint is processed correctly	Passed
Category prediction	Primary category is displayed	Passed
Subcategory prediction	Subcategory is displayed	Passed
Severity prediction	Severity is displayed	Passed
Prediction display	Results are shown through prototype	Passed

The tests indicate that the basic functionality of the prototype works as expected.

7.2 Current Achievements

The main accomplishments of the project are:

- Developed the CivicComp dataset with 60,708 multilingual samples.
- English, Hindi and Hinglish complaints included.
- Developed a hierarchical labeling structure, including category, subcategory and severity.
- Generated 15,000 synthetic complaints using multiple Large Language Models (LLMs)
- Created and compared multiple multilingual transformer models.
- Developed the proposed multi-task framework called HiENBERT.
- Achieved 93.28% and 93.42% accuracy and Macro-F1 on the combined dataset with HiENBERT-XLMR.
- SHAP and t-SNE for interpretability of the models.
- Developed a Streamlit prototype to demonstrate core AI functionality.

7.3 Limitations

The current system has a number of limitations:

- Currently the whole web-based grievance management system is under construction.
- ThThe current prototype is mainly focused on complaint classification and severity prediction.
- Automated real-time routing and resolution are not yet fully implemented.
- Synthetic data may not represent every variation of real citizen complaints.
- Model performance may vary across different languages and complaint categories.

CHAPTER-8
CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

This project proposed a multilingual AI-based framework for automatic classification and severity prediction of civic complaints. The CivicComp dataset and HiENBERT framework facilitate processing of complaints in English, Hindi and Hinglish.

Experimental results show the effectiveness of transformer-based multilingual models, where HiENBERT-XLMR achieves strong performance on the combined dataset. The Streamlit prototype also demonstrates the actual integration of the trained model.

The project lays the foundation for intelligent and multilingual processing of civic complaints and can be integrated in a complete automated complaint management system.

8.2 FUTURE WORK

These are the following improvements planned:

- Full web-based application and back end integration.
- Add complaint storage and tracking database.
- Route complaints to the correct department automatically.
- Real-time prioritization based on predicted severity.
- Add more languages and real world complaints to the dataset
- Enhance model performance on hard and tail categories.
- Improve explainability capabilities for admins and citizens
- Integrate the framework with a complete end to end civic grievance management work flow.

REFERENCES

[1] M. Esperança, D. Freitas, P. V. Paixão, T. A. Marcos, R. A. Martins, and J. C. Ferreira, "Proactive complaint management in public sector informatics using AI: A semantic pattern recognition framework," Applied Sciences, vol. 15, no. 12, Art. no. 6673, 2025, doi: 10.3390/app15126673.

[2] C. Sierra, F. Huanca, M. Torres, and F. Núñez, "An integrated complaint management system based on large language models: Case study in the electric sector," Array, vol. 28, Art. no. 100570, 2025, doi: 10.1016/j.array.2025.100570.

[3] D. Rakhimzhanov, S. Belginova, and D. Yedilkhan, "Automated classification of public transport complaints via text mining using LLMs and embeddings," Information, vol. 16, no. 8, Art. no. 644, 2025, doi: 10.3390/info16080644.

[4] Y. Fang and Z. Shan, "Optimising smart city evaluation: A people-oriented analysis method," IET Smart Cities, vol. 6, no. 1, pp. 41–53, 2024, doi: 10.1049/smc2.12073.

[5] E. Gilman et al., "Addressing data challenges to drive the transformation of smart cities," ACM Transactions on Intelligent Systems and Technology, vol. 15, no. 5, Art. no. 88, 2024, doi: 10.1145/3663482.

[6] A. Singh, R. Bhatia, and S. Saha, "Complaint and severity identification from online financial content," IEEE Transactions on Computational Social Systems, vol. 11, no. 1, pp. 660–670, 2024, doi: 10.1109/TCSS.2022.3215528.

[7] A. Singh, P. Jha, S. Das, R. Jain, and S. Saha, "Toward multimodal complaint severity detection from social media," IEEE Transactions on Computational Social Systems, vol. 11, no. 5, pp. 5903–5912, 2024, doi: 10.1109/TCSS.2024.3393729.

[8] V. Agbesi et al., "Pre-trained transformer-based models for text classification using low-resourced Ewe language," Systems, vol. 12, no. 1, Art. no. 1, 2023, doi: 10.3390/systems12010001.

[9] F. Shama, A. Aziz, and L. B. M. Deya, "CitySolution: A complaining task distributive mobile application for smart city corporation using deep learning," SoftwareX, vol. 27, Art. no. 101829, 2024, doi: 10.1016/j.softx.2024.101829.

[10] A. Yadav et al., “From translation to generative LLMs: Classification of code-mixed affective tasks,” IEEE Transactions on Affective Computing, vol. 16, no. 3, pp. 2090–2101, 2025, doi: 10.1109/TAFFC.2025.3553399.

[11] J. Sterphone, “Complaining by category: Managing social categories and action ascription in wargame interactions,” Language & Communication, vol. 84, pp. 46–60, 2022, doi: 10.1016/j.langcom.2022.03.001.

[12] A. Singh, S. Chandrasekar, T. Sen, and S. Saha, “Federated multitask learning for complaint identification using graph attention network,” IEEE Transactions on Artificial Intelligence, vol. 5, no. 3, pp. 1277–1286, 2024, doi: 10.1109/TAI.2023.3285196.

[13] B. R. Lidiawaty, E. Suryani, and R. A. Vinarti, “Traffic urgency model: A novel approach to prioritize complaint texts using enhanced named entity recognition for Bahasa Indonesia cases,” IEEE Access, vol. 13, pp. 57720–57733, 2025, doi: 10.1109/ACCESS.2025.3556423.

[14] G. Wang, R. Lu, and Y. L. Guan, “Achieve privacy-preserving priority classification on patient health data in remote eHealthcare system,” IEEE Access, vol. 7, pp. 33565–33576, 2019, doi: 10.1109/ACCESS.2019.2891775.

[15] K. Zhang et al., “Heterogeneous multiviews-based efficient graph contrastive learning model for short text classification,” Knowledge-Based Systems, vol. 327, Art. no. 114121, 2025, doi: 10.1016/j.knosys.2025.114121.

[16] A. Salleh et al., “A hybrid model for low-resource language text classification and comparative analysis,” Knowledge-Based Systems, vol. 326, Art. no. 114068, 2025, doi: 10.1016/j.knosys.2025.114068.

[17] R. S. Rao et al., “Implementing NLP to categorize grievances received via a voice input mechanism,” International Journal of Engineering Research & Technology, vol. 12, no. 4, 2023, doi: 10.17577/IJERTV12IS040279.

[18] S. Raj and A. Kumar, “Digital grievance redressal for cleaner, smarter India,” International Journal of Computer Techniques, vol. 12, no. 3, 2025. https://ijctjournal.org/wp-content/uploads/2025/06/DIGITAL-GRIEVANCE-REDRESSAL-FOR-CLEANER-SMARTER-INDIA.pdf

[19] M. Fauzi, “Automatic complaint classification system using classifier ensembles,” Telfor Journal, vol. 10, no. 2, pp. 123–128, 2018, doi: 10.5937/telfor1802123A.

[20] P. Padmane et al., “Multilingual speech and text recognition and translation,” International Journal of Innovations in Engineering and Science, vol. 7, no. 8, pp. 84–88, 2022, doi: 10.46335/IJIES.2022.7.8.15.

[21] P. Gupta et al., “AI-based solution to enable ease of grievance lodging and tracking for citizens across multiple departments,” in Proc. Int. Conf. Advances and Applications in Artificial Intelligence (ICAAAI), 2025, pp. 1002–1022, doi: 10.2991/978-94-6463-738-0_78.

[22] P. Jain, S. Tripathi, T. Garg, N. Sadashiv, N. C. Kundur, M. Phadke, M. Gaikwad, and A. K. Shahade, “An intelligent transformer based framework for bilingual financial complaint classification,” *Scientific Reports*, 2026, doi: 10.1038/s41598-026-51771-w.

[23] Y. Yang, D.-L. Wang, F. Chiclana, and Z.-S. Chen, “A multi-agent LLM framework for severity classification of complaint events: Probabilistic reasoning with scene uncertainty,” *Advanced Engineering Informatics*, vol. 69, Art. no. 104135, 2026, doi: 10.1016/j.aei.2025.104135.

[24] N. Sulaiman and N. Muda, “Automated triaging of hospital complaints using Large Language Model-Assisted Content Analysis and machine learning,” *Engineering Applications of Artificial Intelligence*, vol. 178, Art. no. 114942, 2026, doi: 10.1016/j.engappai.2026.114942.

[25] R. Mishra, A. Kaliappan, S. D. V., T. V. Hyma Lakshmi, P. K. Ravi, Y. Harika Devi, S. Srivastava, D. S. Asudani, S. M. Nagaraj, and Y. H. Bhosale, “Domain-adaptive multitask BERT with graph context modeling for code-mixed Hinglish sentiment classification,” *Engineering, Technology & Applied Science Research*, vol. 16, no. 1, pp. 32336–32341, Feb. 2026, doi: 10.48084/etasr.16589.

[26] S. Mengal, P. Khawale, A. Gade, R. Bhand, and R. B. Diwate, “Community issue prioritization system with ML-based complaint classification,” *International Journal of Creative and Open Research in Engineering and Management*, vol. 2, no. 5, 2026, doi: 10.55041/ijcope.v2i4.1009.

APPENDICES

10.1 Sample Inputs and Outputs



Figure. 10.1: Output By IndicBERT for Primary Category, Secondary Category, Severity Level

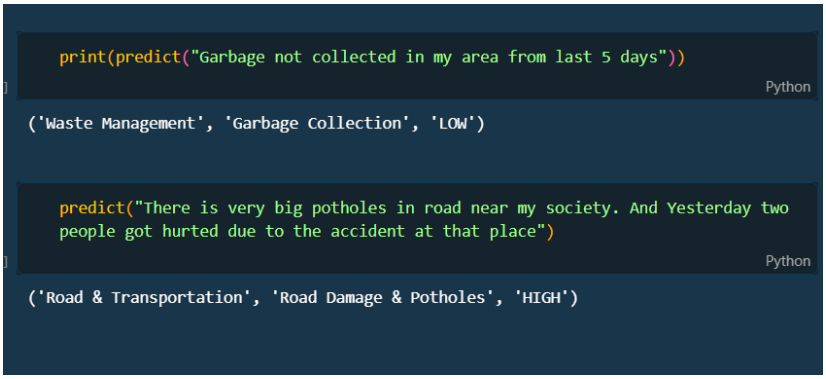


Figure. 10.2: Output By XLM_RoBERTa for Primary Category, Secondary Category, Severity Level

10.2 Additional Graphs and Charts

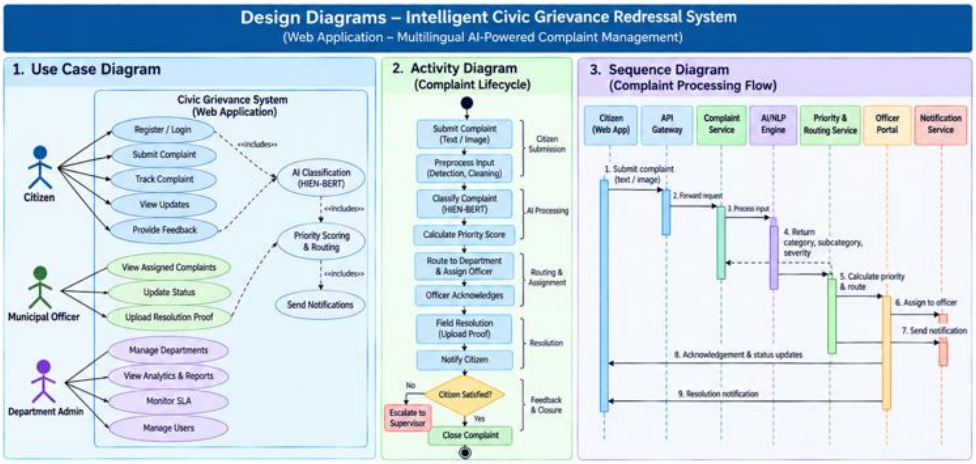
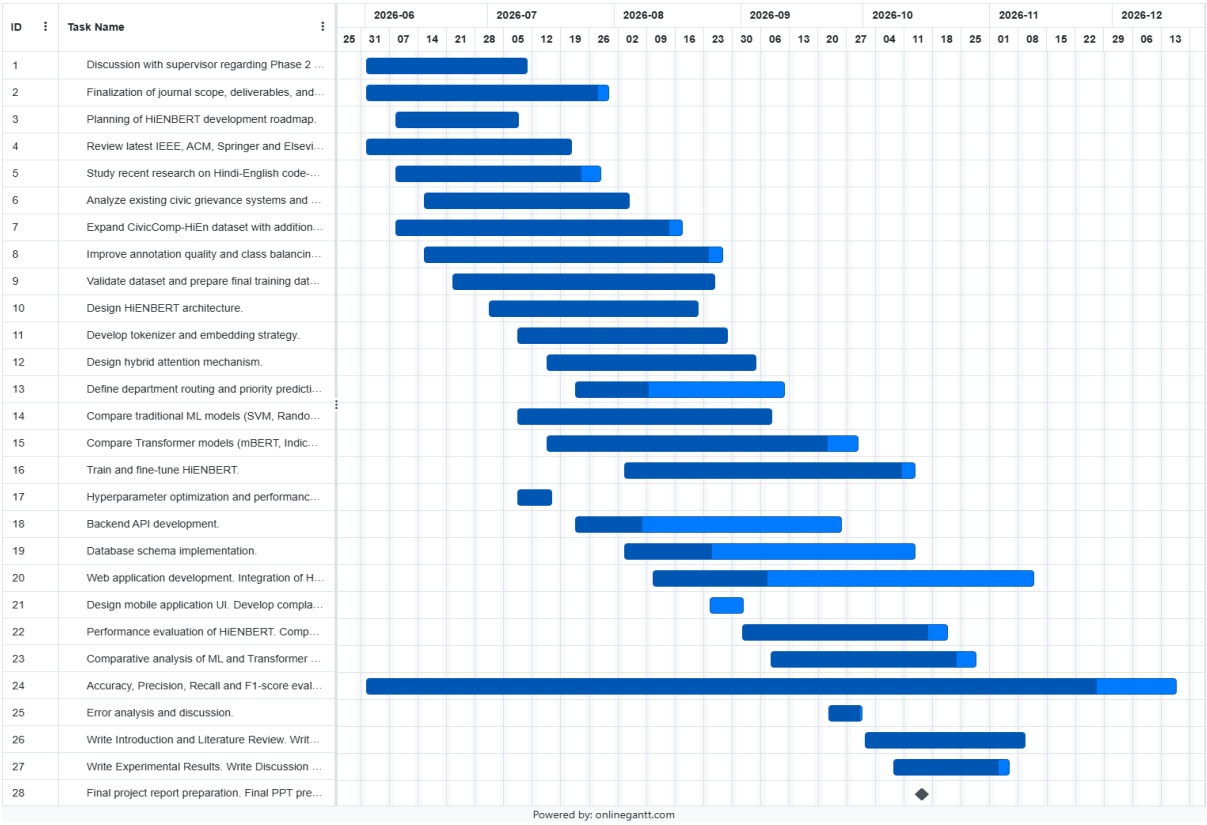
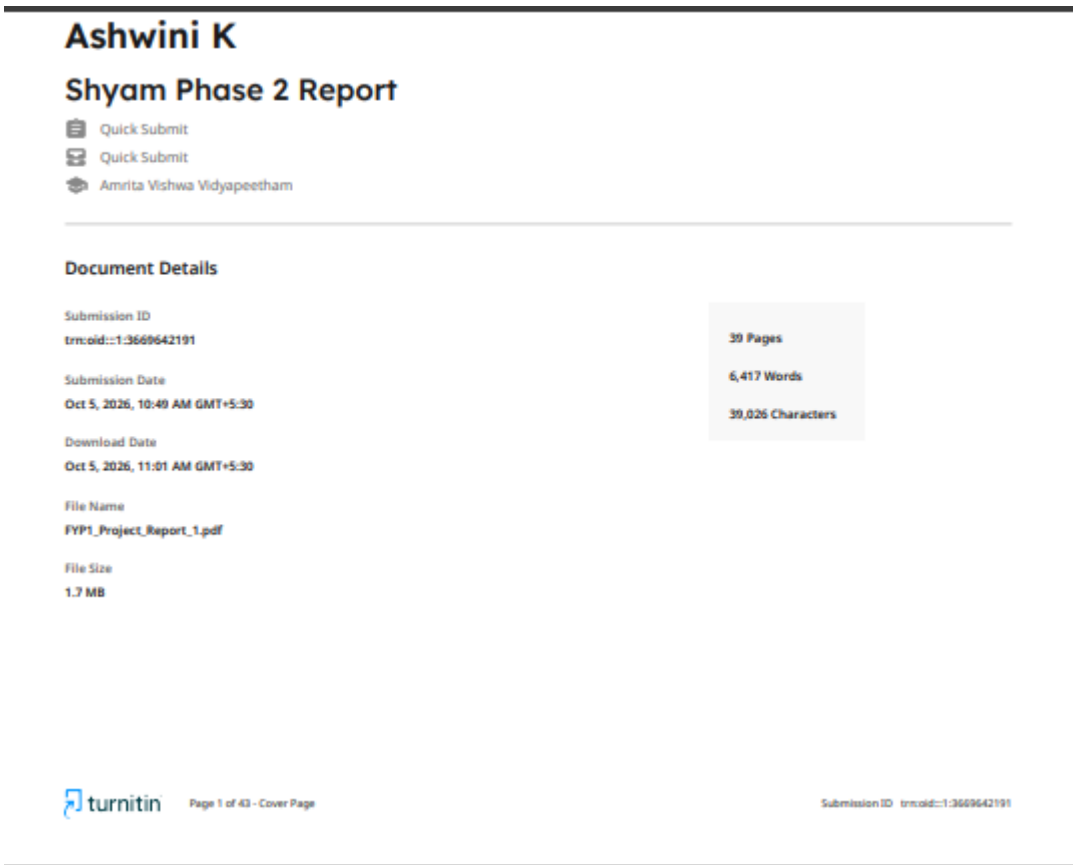


Figure 10.3: Design Diagram

10.3 Gantt Charts



PLAGARISM REPORT



Ashwini K

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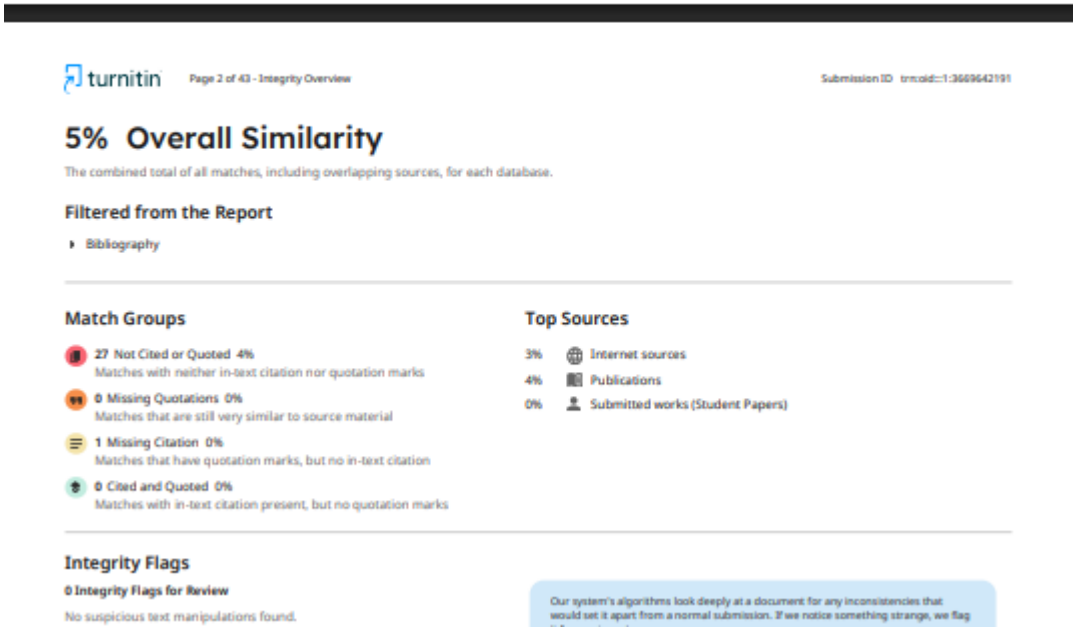
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Shyam Tripathi

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